

MDP

Chance

1. Imagine an MDP M , modelling some non-deterministic as well as probabilistic system. Let s be a state of M . What does $s \models \mathbf{P}_{\geq c}[\phi]$ and $s \models \mathbf{P}_{\leq c}[\phi]$ mean in the context of MDP?

Answer: $s \models \mathbf{P}_{\geq c}[\phi]$ means that the probability that a path/execution that starts from s would behave as ϕ is $\geq c$, *under all possible adversaries*.

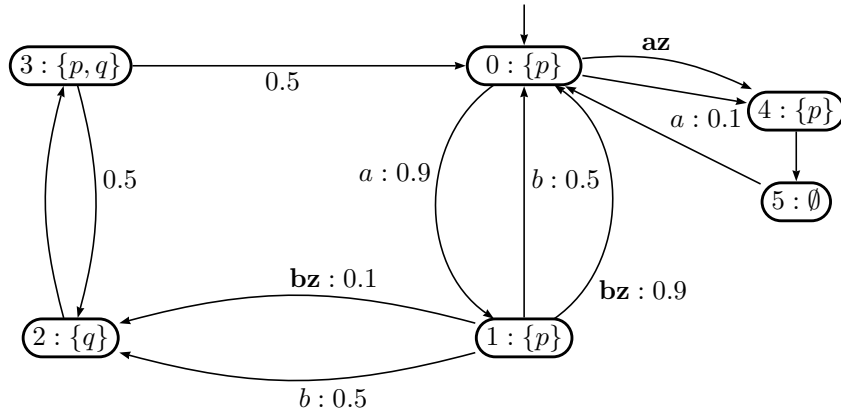
$s \models \mathbf{P}_{\leq c}[\phi]$ is similar: it means that the probability that a path/execution that starts from s would behave as ϕ is $\leq c$, *under all possible adversaries*.

We can note that to check $s \models \mathbf{P}_{\geq c}[\phi]$, we can instead check whether that the *minimum* probability (quantified over all possible adversaries) that a path/execution that starts from s would behave as ϕ is $\geq c$.

Similarly, to check $s \models \mathbf{P}_{\leq c}[\phi]$, we can instead check whether that the *maximum* probability (quantified over all possible adversaries) that a path/execution that starts from s would behave as ϕ is $\leq c$.

2. Consider the following MDP M modelling some non-deterministic as well as probabilistic system. There is non-determinism in states 0 and 1. In states 0, two actions are possible a and $\mathbf{az}$. The choice between them is non-deterministic. The action a leads to either state 1 or 4, but with different probability. The action $\mathbf{az}$ is non-probabilistic: it always lead to state 4.

In state 1, we also have two actions that are possible: b and $\mathbf{bz}$. The choice between them is non-deterministic. Both actions lead to either state 0 or state 2, but with different probabilities.



Describe the procedure to check $0 \models \mathbf{P}_{\geq 0.01} [p \mathbf{U} q]$ on M .

Is $1 \models \mathbf{P}_{\geq 0.01} [p \mathbf{U} q]$ also valid?

Answer: First note that in an MDP the formula $\mathbf{P}_{\geq 0.01} [p \mathbf{U} q]$ that the minimum probability for satisfying $p \mathbf{U} q$, under any adversary, is at least 0.01. We are asked to check if this is true on state 0. To do this will proceed by calculating the minimum probability of the property, for any state, and under any adversary.

- (a) Like in the DTMC case, we will first do some 'pre-computation' to calculate the $\mathbf{S}^{no}$ and $\mathbf{S}^{yes}$ sets¹. But do keep in mind that the interpretation of these sets is a bit (subtly) different:

- i. $\mathbf{S}^{no}$ is the set of states s : there exists an adversary such that $Prob(s, p \mathbf{U} q) \leq 0$ (the probability that there is an execution starting from s and satisfying $p \mathbf{U} q$ would be ≤ 0).
- ii. $\mathbf{S}^{yes}$ is the set of states s satisfying: for all adversaries, $Prob(s, p \mathbf{U} q) \geq 1$ (the probability that there is an execution starting from s and satisfying $p \mathbf{U} q$ is ≥ 1).

To calculate $\mathbf{S}^{no}$, we first calculate the set E of the states satisfying $\mathbf{E}[p \mathbf{U} q]$ under **all** adversaries. We can't literally apply CTL MC to calculate this, because it does not have a concept of adversary. We will use a modified fix-point iteration; see the slides "Model checking MDPs". For convenience it is shown below² The R in the algorithm is our E .

• **Minimum probabilities 0**

$$- \mathbf{S}^{no} = \{ s \in S \mid p_{\min}(s, \phi_1 \mathbf{U} \phi_2) = 0 \} = \text{Sat}(\neg P_{>0}[\phi_1 \mathbf{U} \phi_2])$$

PROB0E($Sat(\phi_1), Sat(\phi_2)$)	
1.	$R := Sat(\phi_2)$
2.	$done := \text{false}$
3.	while ($done = \text{false}$)
4.	$R' := R \cup \{s \in Sat(\phi_1) \mid \forall \mu \in Steps(s). \exists s' \in R. \mu(s') > 0\}$
5.	if ($R' = R$) then $done := \text{true}$
6.	$R := R'$
7.	endwhile
8.	return $S \setminus R$

We start with R_0 to be the states where q holds. So, $R_0 = \{2, 3\}$. In the next iteration R_1 would be R_0 plus all states s where p holds, and where every action α executable on s has at least one transition that leads to the previous R (so, R_0)³ This adds $\{1\}$ to R_1 . Notice that on state 1, **both** actions b and **bz** has a transition that leads to R_0 . So, we have $R_1 = \{1, 2, 3\}$.

In the next iteration we calculate R_2 in the same way. It includes R_1 , plus all states s where p holds, and where every action executable on s has at least one transition that leads to the previous R (so, R_1). The only candidate for such a state is state 0, because it has a direct transition into R_1 . However, not all actions on state 0 leads to R_1 . In particular action **az** has no transition from 0 into R_1 . So we conclude that $R_2 = R_1$. This is a fix point, so we are done with the result $R = \{1, 2, 3\}$.

This R is our E . $\mathbf{S}^{no}$ is the complement of E , so we end up with $\mathbf{S}^{no} = \{0, 4, 5\}$.

We now calculate $\mathbf{S}^{yes}$. Recall that it is the set of all states s satisfying $Prob(s, p \mathbf{U} q) \geq 1$, for *all* (memory-less) adversaries. We first calculate F , the set of states t , from where we have a path whose states satisfies $p \wedge \neg q$, and leads to $\mathbf{S}^{no}$ under *some* adversary. This gives $F = \{0, 1, 4, 5\}$. $\mathbf{S}^{yes}$ is the complement of F , so $\mathbf{S}^{yes} = \{2, 3\}$.

- (b) We can now proceed to calculate the minimum probability $Prob_{\min}(s, p \mathbf{U} q)$ for every state, regardless the adversary (under any adversary).

¹There is a note in the slides "Model checking MDPs" that calculating $\mathbf{S}^{yes}$ is not required, though it speeds up the checking.

²Note that the algorithm is for computing E and $\mathbf{S}^{no}$ for the 'min' case (for calculating minimum probabilities). For the 'max' case (calculating maximum probabilities), $\mathbf{S}^{no}$ and $\mathbf{S}^{yes}$ have subtly different interpretation. See the slides "Model checking MDPs".

³Note that it is not necessary that all α -transitions leads to R_0 ; one is enough – but this has to hold for all actions $\alpha, \beta, \dots$ executable on s .

For every state s , let's introduce x_i representing the value of $Prob_{min}(s, p \text{ U } q)$ regardless the adversary.

For members of $\mathbf{S}^{yes}$ we know, by its definition, that their x_s must be 1. For every member s of $\mathbf{S}^{no}$, we know that for some adversary $Prob(s, p \text{ U } q) \geq 0$. So this 0 is also the minimum over all adversaries. So, $x_s = 0$.

This means that these are known: $x_2 = x_3 = 1$ And $x_0, x_4, x_5 = 0$.

We can now answer the first of the initial questions, namely whether $0 \models \mathbf{P}_{\geq 0.01} [p \text{ U } q]$ holds. It does not, because $x_0 = 0$.

For the second question we have to know what x_1 is. This remains to be calculated. I will do this by converting the problem to a Linear Programming (LP) problem. On state 1, two actions are possible, b and $\mathbf{bz}$. These generate the following constraints for x_1 :

$$\begin{aligned} x_1 &\leq P_{b:1 \rightarrow 0} * x_0 + P_{b:0 \rightarrow 2} * x_2 && \text{(coming from } b) \\ x_1 &\leq P_{\mathbf{bz}:1 \rightarrow 0} * x_0 + P_{\mathbf{bz}:0 \rightarrow 2} * x_2 && \text{coming from } \mathbf{bz}) \end{aligned}$$

The solution for x_1 is to be found by finding the largest x_1 that meets the above constraints⁴

I will first work out the constraints. Since $x_0 = 0$ and $x_2 = 1$, we then get:

$$\begin{aligned} x_1 &\leq P_{b:0 \rightarrow 2} \\ x_1 &\leq P_{\mathbf{bz}:0 \rightarrow 2} \end{aligned}$$

Filling in the numbers:

$$\begin{aligned} x_1 &\leq 0.5 \\ x_1 &\leq 0.1 \end{aligned}$$

We should maximize x_1 while satisfying the constraints above. In this case the solution is easy, namely $x_1 = 0.1$.

The the second question, we have to check whether $1 \models \mathbf{P}_{\geq 0.01} [p \text{ U } q]$ holds. Well, it does because $x_1 = 0.1$.

3. Consider again the model in No. 2. Check if this is valid: $0 \models \mathbf{P}_{\leq 0.82} [p \text{ U } q]$. Show how the model check works for this case.

Answer: Just mentioning some key differences here. The definition of $\mathbf{S}^{no}$ and $\mathbf{S}^{yes}$ is subtly different than the 'min' case.

- (a) $\mathbf{S}^{no}$ is the set of states s : for *all* adversaries, $Prob(s, p \text{ U } q) \leq 0$.
- (b) $\mathbf{S}^{yes}$ is the set of states s satisfying: there exists an adversary, such that $Prob(s, p \text{ U } q) \geq 1$.

See the slides "Model checking MDPs" for the algorithm to calculate $\mathbf{S}^{no}$ for the 'max' case.

⁴In this example, maximizing only x_1 works and sensical (because other x_s is already fixed). In a different setup, if for example x_t and x_u are still unknown, then we should find solutions that maximize $x_t + x_u$ while fulfilling constraints such as those above for x_1 .